23/09/2026

Code-C
Phase-3

Aakash
Medical | IIT-JEE | Foundations

***Corporate Office*:** AESL, 3rd Floor, Incuspaze Campus-2, Plot no. 13, Sector-18, Udyog Vihar, Gurugram, Haryana-122015

FORTNIGHTLY TEST FOR NEET-2027 (XII-PASSED)

MM: 720 **FT-03** **Time: 180 Mins.**

Answers

1. (2)	37. (2)	73. (4)	109. (3)	145. (2)
2. (2)	38. (1)	74. (2)	110. (3)	146. (1)
3. (1)	39. (4)	75. (1)	111. (4)	147. (3)
4. (3)	40. (2)	76. (4)	112. (4)	148. (1)
5. (3)	41. (4)	77. (3)	113. (3)	149. (1)
6. (2)	42. (2)	78. (3)	114. (2)	150. (1)
7. (2)	43. (2)	79. (3)	115. (1)	151. (2)
8. (2)	44. (3)	80. (3)	116. (3)	152. (3)
9. (1)	45. (3)	81. (3)	117. (3)	153. (1)
10. (1)	46. (2)	82. (2)	118. (1)	154. (1)
11. (2)	47. (4)	83. (3)	119. (2)	155. (4)
12. (2)	48. (2)	84. (4)	120. (3)	156. (2)
13. (2)	49. (1)	85. (1)	121. (4)	157. (4)
14. (2)	50. (4)	86. (4)	122. (4)	158. (3)
15. (1)	51. (2)	87. (4)	123. (3)	159. (3)
16. (4)	52. (3)	88. (1)	124. (2)	160. (4)
17. (3)	53. (2)	89. (3)	125. (2)	161. (4)
18. (2)	54. (4)	90. (4)	126. (4)	162. (3)
19. (3)	55. (4)	91. (2)	127. (3)	163. (1)
20. (3)	56. (2)	92. (3)	128. (3)	164. (2)
21. (1)	57. (1)	93. (4)	129. (1)	165. (2)
22. (4)	58. (4)	94. (1)	130. (1)	166. (4)
23. (2)	59. (1)	95. (3)	131. (2)	167. (2)
24. (4)	60. (4)	96. (3)	132. (2)	168. (3)
25. (2)	61. (2)	97. (3)	133. (3)	169. (4)
26. (3)	62. (4)	98. (3)	134. (4)	170. (1)
27. (3)	63. (1)	99. (3)	135. (2)	171. (3)
28. (1)	64. (2)	100. (3)	136. (1)	172. (4)
29. (1)	65. (4)	101. (2)	137. (1)	173. (3)
30. (3)	66. (3)	102. (3)	138. (1)	174. (2)
31. (1)	67. (2)	103. (2)	139. (1)	175. (1)
32. (3)	68. (1)	104. (2)	140. (2)	176. (2)
33. (2)	69. (4)	105. (4)	141. (3)	177. (2)
34. (3)	70. (2)	106. (1)	142. (1)	178. (3)
35. (3)	71. (4)	107. (1)	143. (3)	179. (2)
36. (1)	72. (1)	108. (2)	144. (2)	180. (2)

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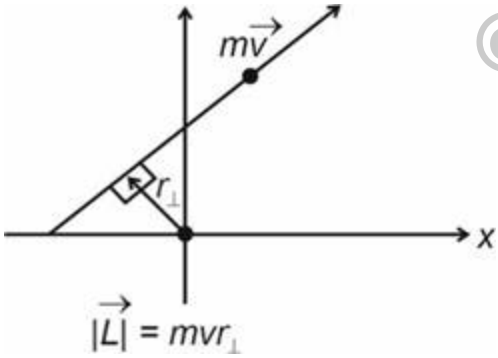
Hints and Solutions

PHYSICS

(1) **Answer :** (2)
Solution:
Gravitational force is a conservative force and centre of mass of a body may lie outside the body.

(2) **Answer :** (2)
Solution:
At lower most position
$\frac{v^2}{l} = 5g$
$T = 6\ mg$
At upper most position
$T = 0$
$\frac{v^2}{l} = g$
At horizontal position
$T = 3\ mg$
Radial acceleration at uppermost position $mg = ma \Rightarrow a = g$
Radial acceleration at lowermost position $T - mg = ma$
$6mg - mg = ma$
$a = 5g$

(3) **Answer :** (1)
Hint:
Use $L = mvr\sin\theta = mvr_\perp$
Solution:



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For constant velocity, $r_\perp$ remains constant so angular momentum is constant.

(4) **Answer :** (3)
Hint:
We know, $\tau = \frac{dL}{dt}$
Solution:
Statement I is true as torque is the turning effect of force.
Statement II is wrong as $\tau = \frac{dL}{dt} = \frac{d}{dt}(I\omega)$

(5) **Answer :** (3)
Solution:
$P = \frac{dK}{dt}$

(6) **Answer :** (2)
Solution:
For elastic collision $e = 1$

(7) **Answer :** (2)
Solution:

$\vec{r}_{cm}$ =The position vector from point *O* to c.o.m. of rod = zero

$\vec{L}_0 = \vec{L}_{cm} + \vec{r}_{cm} \times m\vec{v}_{cm}$

$= I_{cm}\vec{\omega} + zero = \frac{ml^2}{12}\omega\hat{k}$

(8) **Answer :** (2)
Solution:
By applying conservation of energy

$mgR = \frac{1}{2}mv^2$

$v = \sqrt{2Rg}$

(9) **Answer :** (1)
Solution:

$\vec{P} = \vec{F} \cdot \vec{v}$

$= \left(4\hat{i} + \hat{j} - 2\hat{k}\right)\left(2\hat{i} + 2\hat{j} + 3\hat{k}\right)$ W

= [8 + 2 – 6] W

= 4 W

(10) **Answer :** (1)
Solution:
Constant speed means uniform circular motion
Only centripetal force acts perpendicular to displacement, so no work.

(11) **Answer :** (2)
Solution:
Friction force is not associated with potential energy, so it is a non-conservative force.

(12) **Answer :** (2)
Solution:

$p = \sqrt{2mk}$

$\frac{p_1}{p_2} = \frac{\sqrt{2m_1k_1}}{\sqrt{2m_2k_2}}$

$\frac{p_1}{p_2} = \frac{3}{2}$

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(13) **Answer :** (2)
Hint:

$<E> = \frac{\int (E)dt}{\int dt}$

Solution:

$<E>_{t=0 \text{ to } 2\,\text{s}} = \frac{\int_0^2 4t^3 dt}{\int_0^2 dt} = \frac{\left[t^4\right]_0^2}{2-0}$

$= \frac{2^4-0}{2-0} = 8$ J

(14) **Answer :** (2)
Solution:
According to Newton's 3rd law, force exerted by body *A* on *B* is equal to force exerted by body *B* on body *A* but not true for work done.

(15) **Answer :** (1)
Solution:

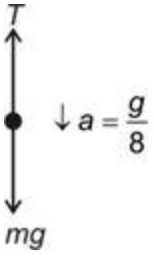
$F = -\frac{\partial U}{\partial x} < 0$

Hence force is along negative *x*-axis.

(16) **Answer :** (4)
Hint:

Work done = $\vec{F} \cdot \vec{d}$

Solution:



$T = mg - \frac{mg}{8} = \frac{7mg}{8}$

Work done by the cord $W = T \times d\cos 180°$

$= -\frac{7mg}{8} \cdot d = -\frac{7mgd}{8}$

(17) **Answer :** (3)
Hint:

Energy stored in stretched spring $U = \frac{1}{2}kx^2 = \frac{F^2}{2k}$

Solution:

Given $K_A = 3K_B$

$\frac{U_A}{U_B} = \frac{K_B}{K_A}$

$\frac{E}{E_B} = \frac{K_B}{3K_B}$

$E_B = 3E$

(18) **Answer :** (2)
Solution:

Since the ball after collision goes upto a height of $\frac{h}{4}$ only, hence percentage loss in energy would be 75%.

(19) **Answer :** (3)
Solution:

Conservation of linear momentum holds true for all type of collision.

(20) **Answer :** (3)
Solution:

$W = FS\cos\theta$

$= 50 \times 10 \times \cos 60^\circ = \frac{500}{2} = 250\ \text{J}$

(21) **Answer :** (1)
Solution:

Work = Area under ($F - d$) graph

= 8 + 5 = 13 J

(22) **Answer :** (4)
Solution:

$-\frac{dU}{dr} = 0 \Rightarrow\ r = \frac{2A}{B}$

(23) **Answer :** (2)
Solution:

In elastic collision, velocity of masses get interchanged if masses are equal.

(24) **Answer :** (4)
Solution:

$\text{KE}_{\min} = \frac{1}{2}mv^2 = \frac{1}{2}m(\sqrt{5gl})^2 = \frac{5}{2}mgl$

(25) **Answer :** (2)
Solution:

$P = \frac{dW}{dt}$

$dW = P \cdot dt$

Energy = Power × time

(26) **Answer :** (3)
Solution:

ΔK.E. = Work done

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So,
Work done = $\frac{1}{2} \times 2(3)^2 = 9$ J

(27) **Answer :** (3)
Solution:
KE = $\frac{1}{2}m\left|\vec{v}\right|^2$
= $\mathrm{KE} = \frac{1}{2} \times 1\left(4+16\right)$ = 10 J

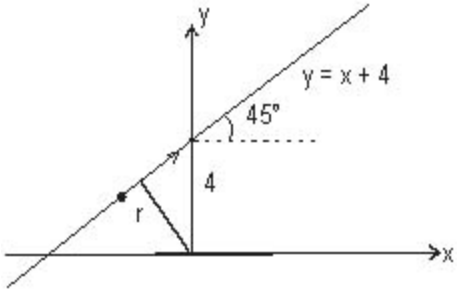
(28) **Answer :** (1)
Solution:
For stable equilibrium, $\frac{dF}{dx} < 0$
So, $x = -x_1$ is position of stable equilibrium.

(29) **Answer :** (1)
Solution:
$\theta = (4t - 3t^2 + t^3)$
$\frac{\mathrm{d}\theta}{\mathrm{dt}} = 4 - 6\mathrm{t} + 3\mathrm{t}^2$
ω(4) = 28 rad/s
ω(2) = 4 rad/s
$\alpha_{\mathrm{av}} = \frac{\omega(4)-\omega(2)}{4-2}$
$\alpha_{avg} = \frac{28-4}{2} = 12\ \mathrm{rad}/\mathrm{s}^2$

(30) **Answer :** (3)
Solution:
$\vec{L}$ = Constant
$\vec{\tau} = 0$

(31) **Answer :** (1)
Solution:
$I = m\left(\frac{d}{2}\right)^2 \times 2$
$= \frac{md^2}{2}$

(32) **Answer :** (3)
Solution:
clearly
r = $\frac{4}{\sqrt{2}}$



angular momentum about origin
L_0 = mvr
= 5 × 3 $\sqrt{2} \times \frac{4}{\sqrt{2}}$
= 60 gcm^2/s

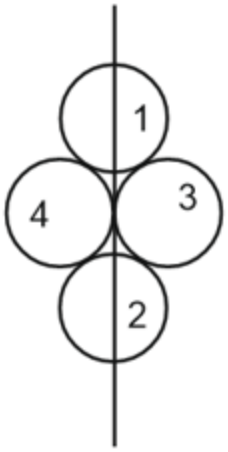
(33) **Answer :** (2)
Solution:
$\vec{\tau} = \frac{d\vec{L}}{dt}$
$d\vec{L} = \vec{\tau}.dt = 20 \times 2 = 40\ \mathrm{kg}\frac{\mathrm{m}^2}{\mathrm{s}}$

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(34) **Answer :** (3)
Solution:
$\Delta L = I(\omega - \omega_0)$
$= \frac{5}{3} \times 2\pi \frac{(120-60)}{60}$
$= \frac{10\pi}{3}$ kg m^2/s

(35) **Answer :** (3)
Solution:
$I = mr^2$
$L = 2\pi r$
$\therefore \ r = \frac{L}{2\pi}$
$I = \frac{M}{2}\left(\frac{L}{2\pi}\right)^2 = \frac{ML^2}{8\pi^2}$

(36) **Answer :** (1)
Solution:



$I_1 = I_2 = \frac{MR^2}{4}$
$I_3 = I_4 = \frac{MR^2}{4} + MR^2$
$= \frac{5}{4}MR^2$
$I_{Axis} = I_1 + I_2 + I_3 + I_4$
$= \frac{MR^2}{2} + \frac{5}{2}MR^2$
$= 3MR^2$

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(37) **Answer :** (2)
Solution:
If torque is zero, then angular momentum is conserved.
If $\vec{F} \| \vec{r}$, then $\tau = 0$
$\therefore \ \frac{\alpha}{4} = \frac{-2}{-4} = \frac{3}{6}$
$\frac{\alpha}{4} = \frac{1}{2}$
$\alpha = \frac{4}{2} = 2$

(38) **Answer :** (1)
Solution:
$P = \tau \cdot \omega$
$P = 10 \times 10 = 100$ W

(39) **Answer :** (4)
Solution:
Torque is given by $\vec{\tau} = \vec{r} \times \vec{F}$

(40) **Answer :** (2)
Solution:

Here α = π rad/s^2

$\omega_0 = 0,\ \ t$ = 3s

$\omega_0 = \omega_0 + \alpha t = 0 + \pi \times 3 = 3\pi$ rad s^{-1}

(41) **Answer :** (4)

Solution:

$\left(2\hat{i} + 4\hat{j}\right) \times \left(\hat{i} + 2\hat{j}\right)$

$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 4 & 0 \\ 1 & 2 & 0 \end{vmatrix}$

= $\hat{i}\left(0\right) + \hat{j}\left(0\right) + \hat{k}\left(4 - 4\right)$

= Zero

(42) **Answer :** (2)

Solution:

Moment of inertia of ring about diameter

= $\frac{MR^2}{2} = I$

Angular momentum $L = I\omega$

$= \frac{MR^2}{2}\omega = 4 \times \frac{(0.25)^2}{2} \times 1000$

$= \frac{1000}{8}$ = 125 J s

(43) **Answer :** (2)

Solution:

$r_{com} = \frac{1(1,0,1) + 1(1,1,0)}{2}$

$= \left(1, \frac{1}{2}, \frac{1}{2}\right)$

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(44) **Answer :** (3)

Hint:

$\vec{v}_{cm} = \frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1 + m_2}$

Solution:

$\vec{v}_{cm} = \frac{2m\hat{i} + 2m\hat{j}}{2m} = \left(\hat{i} + \hat{j}\right)$ m s^{-1}

$\left|\vec{v}_{cm}\right| = \sqrt{2}$ ms^{-1}

(45) **Answer :** (3)

Solution:

$2FR = \frac{1}{2}MR^2\alpha$

$\alpha = \frac{4F}{MR}$

CHEMISTRY

(46) **Answer :** (2)

Hint:

Increase in disorderness leads to increase in entropy.

Solution:

Entropy increases during boiling of egg because of denaturation of protein.

$H_2(g) \rightarrow 2H(g)$

Entropy increases because number of gaseous moles increases.

(47) **Answer :** (4)

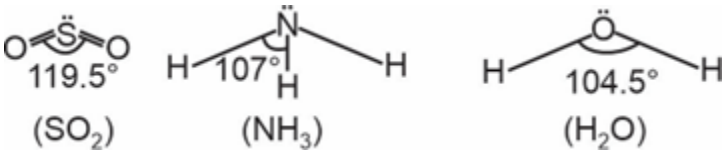
Solution:

1st law of thermodynamics,

$\Delta U = Q + W = 100\ J + (-60\ J) = 40\ J$

(48) **Answer :** (2)
Solution:
For reaction
$\Delta_r H = \sum (\Delta_f H)_{Product} - \sum (\Delta_f H)_{Reactant}$
$\Delta_r H = -630 - 400 + 1200$
$= +170\ kJ\ mol^{-1}$

(49) **Answer :** (1)
Solution:



(SO_2) (NH_3) (H_2O)

Species	Bond Order
O_2^+	2.5
N_2	3.0
F_2	1.0

- $H_2(g) \longrightarrow H(g) + H(g);\ \Delta_a H^{\ominus} = 435.8\ kJ/mol$
- $N_2(g) \longrightarrow N(g) + N(g);\ \Delta_a H^{\ominus} = 946.0\ kJ/mol$
- $HCl \longrightarrow H(g) + Cl(g);\ \Delta_a H^{\ominus} = 431.0\ kJ/mol$

(50) **Answer :** (4)
Solution:
• Boiling point of H_2O is greater than H_2S because H_2O is associated by intermolecular hydrogen bonding
• Covalent bond is stronger than hydrogen bond

(51) **Answer :** (2)
Hint:
$\Delta S = \frac{q_{rev}}{T}$
Solution:
For vaporisation
$\Delta S = \frac{\Delta_{vap} H^{\circ}}{T} = \frac{0.5 \times 40.66 \times 10^3}{373} = 54.5\ JK^{-1}$

(52) **Answer :** (3)
Solution:
$C_P - C_V = R$

(53) **Answer :** (2)
Solution:
Percentage ionic character = $\frac{\mu_{observed} \times 100}{\mu_{calculated}}$
$\mu_{observed} = 2.5 \times 3.33 \times 10^{-30}$ C m
$\mu_{calculated} = 1.6 \times 10^{-19} \times 125 \times 10^{-12}$ C m
% ionic character
$= \frac{2.5 \times 3.33 \times 10^{-30} \times 100}{1.6 \times 125 \times 10^{-31}} = 41.6\%$

(54) **Answer :** (4)
Solution:
a. Isothermal process ⇒ Temperature is constant throughout the process
b. Isochoric process ⇒ Volume is constant throughout the process
c. Isobaric process ⇒ Pressure is constant throughout the process
d. Adiabatic process ⇒ No exchange of heat (q) between system and surrounding

(55) **Answer :** (4)

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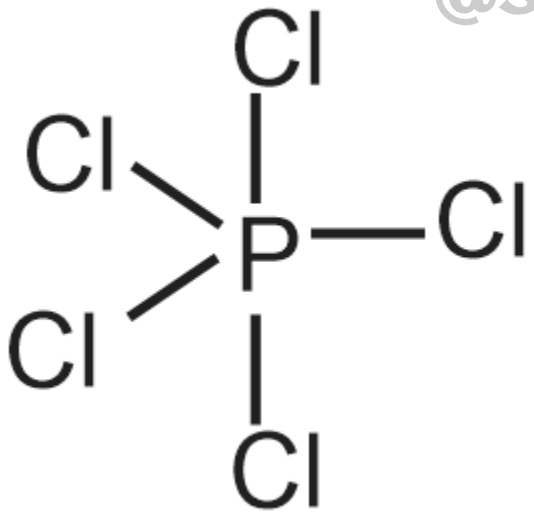
Solution:
No work is done during the free expansion of an ideal gas

(56) **Answer :** (2)
Solution:
Bomb calorimeter is used to measure change in internal energy

(57) **Answer :** (1)
Solution:
$\mathrm{Ne}_2 \ : \ \sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 \ < \ \sigma 2p_z^2 \ < \ (\pi 2p_x^2 \ = \ \pi 2p_y^2)$
$< \ \pi^* 2p_x^2 = \pi^* 2p_y^2 < \ \sigma^* \ 2p_z^2$
$\sigma^* 2p_z \ \text{i.e.} \ \sigma^* 2p$ is highest occupied orbital.

(58) **Answer :** (4)
Hint:
For any reaction
$\Delta_r H = \sum (BE)_{Reactant} - \sum (BE)_{Product}$
Solution:
If BE of X_2 = x
BE of Y_2 = 2x
BE of XY = 3x
$\Delta H = (BE)_{X_2} + (BE)_{Y_2} - 2(BE)_{XY}$
= x + 2x –2(3x)
–400 = –3x
$x = \frac{400}{3}$
BE of Y_2 = 2x = $2 \times \frac{400}{3} = \frac{800}{3} \ \mathrm{kJ/mol}$

(59) **Answer :** (1)
Solution:



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The maximum number of atoms lying in one plane is 4.

(60) **Answer :** (4)
Solution:
Heat is a path function.

(61) **Answer :** (2)
Solution:
• $\Delta H° = \Delta U° + \Delta n_g RT$
= – 10 + (0)RT = –10 kJ
$\Delta G° = \Delta H° - T\Delta S°$
• $\Delta G° = (– 10 – (298 \times 10^{-3} \times (– 40) = 1.92$ kJ

(62) **Answer :** (4)
Solution:
Pressure is an intensive property.

(63) **Answer :** (1)
Solution:
For reversible expansion
$w = -2.303\ nRT \log \frac{V_2}{V_1}$
$w = -2.303 \times 10 \times 8.314 \times 300 \log \frac{240}{12}$
$= -74733\ J$
$= -74.73\ kJ$

(64) **Answer :** (2)
Hint:
at equilibrium $\Delta G = 0$ hence $\Delta G^\circ = -2.303\ RT\ \log K_c$
Solution:
$-9.2 \times 10^3 = -2.303 \times 2 \times 400\ \log K_c$
$\log K_c = \frac{9.2 \times 1000}{2.303 \times 2 \times 400}$
$= 5$
$K_c = 10^5$

(65) **Answer :** (4)
Solution:
Covalent character directly proportional to larger size of anion. (Fajan's Rules)
CaI_2 has maximum covalent character.

(66) **Answer :** (3)
Solution:
2 unpaired electrons are present in B_2

(67) **Answer :** (2)
Hint:
$C_{(graphite)} + O_2(g) \rightarrow CO_2(g);\ \Delta_f H$

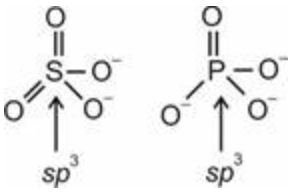
$\Delta_f H = -393 kJ\ mol^{-1}$
Solution:
Heat release upon combustion of 1 mole of carbon (graphite) = $\Delta_f H\ (CO_2)$
Since, upon of formation of 44 g of CO_2, 393 kJ heat is released
Therefore, heat released upon formation of 27.5 g of $CO_2 = \frac{393}{44} \times 27.5 kJ$
= 245.63 g

(68) **Answer :** (1)
Solution:
Since NaOH and HCl both are strong electrolyte it will give maximum heat of neutralisation.

(69) **Answer :** (4)
Hint:
The process is adiabatic hence q = 0
Solution:
$w = -P\Delta V$
$= -4 \times (5.75 - 1.25)\ L\ atm$
$= -18\ L\ atm$
$w = -18 \times 101.3 = -1.82\ kJ$
From first law, $\Delta U = q + w$
$\Delta U = w = -1.82\ kJ$

(70) **Answer :** (2)
Hint:
For sp^3 hybridisation the bond pair/lone pair should be arranged tetrahedrally around the central atom.
Solution:

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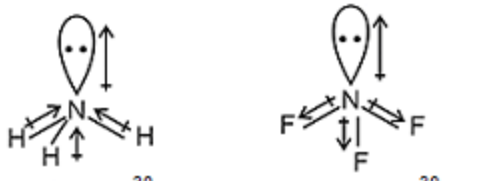
(71) **Answer :** (4)
Solution:
We know,
$\Delta S = 2.303\ nR\ \log \frac{V_2}{V_1}$
Given: $V_2 = 20$ L ; $V_1 = 2$ L
n = 2 mol., R = 2 cal mol^{-1} K^{-1}
$\therefore \Delta S = 2.303 \times 2\ mol \times 2\ cal\ mol^{-1}\ K^{-1} \times \log \frac{20}{2} = 9.2\ cal\ K^{-1}$

(72) **Answer :** (1)
Solution:
$q = mS\Delta T$
$= 50 \times 0.39 \times (80 - 35)$
$q = 877.5$ J

(73) **Answer :** (4)
Hint:
For a given cation, smaller is the size of anion more is the lattice enthalpy.
Solution:
Correct order of lattice enthalpy
NaF > NaCl > NaBr > NaI

(74) **Answer :** (2)
Solution:
O_2^+ : $\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_z^2, \pi 2p_x^2, \pi 2p_y^2, \pi^* 2p_x^1$

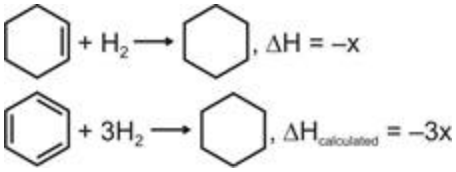
(75) **Answer :** (1)
Solution:



$\mu_{net} = 4.9 \times 10^{-30}$ Cm $\quad \mu_{net} = 0.8 \times 10^{-30}$ Cm

(76) **Answer :** (4)
Solution:
$\Delta H = \Delta U + \Delta n_g RT$
For, $CH_4\ (g) + 2O_2\ (g) \rightarrow CO_2\ (g) + 2H_2O\ (l)$
$\Delta n_g = (1-3) = -2$
So, $\Delta H \neq \Delta U$

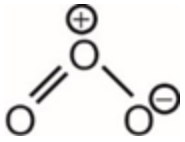
(77) **Answer :** (3)
Hint:
Resonance energy
$(RE) = (\Delta H_f)_{calculated} - (\Delta H_f)_{experimental}$
Solution:



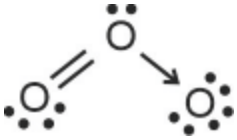
$\therefore$ RE = (–3x) – (–y) = y – 3x.

(78) **Answer :** (3)
Solution:

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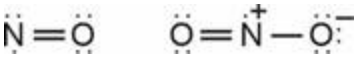
or



$\mathrm{F.C} = 6 - \frac{1}{2} \times 6 - 2 = 6 - 5 = +1$

(79) **Answer :** (3)
Hint:
Electrons are filled in the orbitals on the basis of increasing energy level.
Solution:
$\mathrm{F_2} = \sigma 1s^2, \sigma * 1s^2, \sigma 2s^2, \sigma * 2s^2, \sigma 2p_z^2$
$\pi 2p_x^2 = \pi 2p_y^2 \ \pi * 2p_x^2 \ = \pi * 2p_y^2$
$\mathrm{N_2} = \sigma 1s^2 \ \sigma * 1s^2 \ \sigma 2s^2 \ \sigma * 2s^2 \ \pi 2p_x^2 = \ \pi p_y^2 \ \sigma 2p_z^2$

(80) **Answer :** (3)
Hint:
Molecules which have an odd number of electrons are called odd-electron molecules.
Solution:
Molecules like NO and NO_2 are odd-electron molecules.

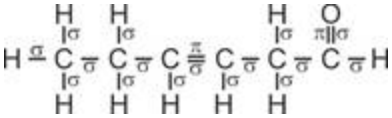


(81) **Answer :** (3)
Solution:
• More is the bond order, more is the stability.
• Also, stability decreases as the number of electrons in antibonding molecular orbital increases.
∴ Order of stability is
$\mathrm{N_2} > \mathrm{N_2^+} > \mathrm{N_2^-}$

(82) **Answer :** (2)
Hint:
If number of bonding and antibonding electrons are equal then species does not exist.
Solution:
Be_2 has equal bonding and antibonding electrons, so does not exist.

(83) **Answer :** (3)

(84) **Answer :** (4)
Hint:
Double bond has 1σ and 1π bond.
Solution:



Number of σ bonds = 16
Number of π bonds = 2

(85) **Answer :** (1)
Hint:
The standard molar enthalpy of formation at 298 K for C(graphite) is zero
Solution:

Substance	Δ_f H°(kJ mol^{-1}
$\mathrm{Br_2(l)}$	0
H_2(g)	0
HI(g)	+26.48
Cl_2(g)	0

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C(diamond)	+1.89
S(rhombic)	0
Si(s)	0
HCl(g)	–92.31

(86) **Answer :** (4)
Solution:
$\frac{1}{2}X_2(g) + Y_2(g) \rightarrow XY_Z(g)$
$\Delta S^\circ_{\text{ramoton}} = \Delta S^\circ_{\text{Pr oduat}} - \Delta S^\circ_{\text{ramotznt}}$
$-50 - \frac{1}{2} \times 60 - 40$
= – 20 J K^{-1} mol^{-1}
∴ ΔG = ΔH – TΔS
At equilibrium, ΔG = 0
$T = \frac{\Delta H}{\Delta S} = \frac{-20 \times 10^2}{-20} = 1000\ \text{K}$

(87) **Answer :** (4)

(88) **Answer :** (1)
Solution:
$\Delta \mathrm{T} = 0\ \&\ \Delta\,\mathrm{U} = \mathrm{nC_V}\,\Delta\,\mathrm{T}$

$\therefore \Delta \mathrm{U} = 0$

(89) **Answer :** (3)
Solution:
O_2 = Bond order = 2
O_2^+ = Bond order = 2.5
O_2^- = Bond order = 1.5
O_2^{2-} = Bond order = 1

(90) **Answer :** (4)
Hint: If ΔG is +ve, reaction is always non-spontaneous
Solution:
ΔG = ΔH – TΔS
If ΔH is +ve and ΔS is –ve then
ΔG is always +ve at all temperatures.

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BOTANY

(91) **Answer :** (2)
Hint:
Ray florets of sunflower are epigynous flowers.
Solution:
Ray florets of sunflower have inferior ovary.

(92) **Answer :** (3)
Hint:
Floral formula does not represent placentation.
Solution:
In case of pea, condition of stamens is diadelphous, *i.e.*, 9 + 1 type.

(93) **Answer :** (4)
Hint:
This type of leaf is palmately compound leaf.
Solution:
In palmately compound leaves, the leaflets are attached at a common point, i.e., the tip of petiole, as in silk cotton.

(94) **Answer :** (1)
Solution:
Rose flower have half-inferior ovary, thus it is known as perigynous flower.

In pea, the placenta form a ridge along the ventral suture of the ovary and ovules are borne on this ridge forming two rows.
In cotton, twisted aestivation is present.
In Mango, fruit is drupe.

(95) **Answer :** (3)
Hint:
Aggregate fruits are developed from the polycarpellary and apocarpous ovary.
Solution:
If a fruit is formed without fertilisation it is called a parthenocarpic fruit.

(96) **Answer :** (3)
Hint:
In bean flower, the innermost petal is keel, which is also known as carina.
Solution:
In bean flowers, among the five petals, the largest (standard) overlaps the two lateral petals (wings) which in turn overlap the two smallest anterior petals (keel).

(97) **Answer :** (3)
Solution:
Staminode – Sterile stamen
Syncarpous – Fused carpels
Gamopetalous – Petals united
Polysepalous – Sepals free

(98) **Answer :** (3)
Solution:
Epipetalous condition of stamens is found in family Solanaceae.

(99) **Answer :** (3)
Hint:
Mesocarp of mango is fleshy.
Solution:
Mesocarp of coconut is fibrous, endocarp is hard and stony.

(100) **Answer :** (3)
Solution:
In monocotyledons, leaf base expands into a sheath covering the stem partially or wholly.

(101) **Answer :** (2)
Solution:
In compound leaf the thickest midrib is called rachis.

(102) **Answer :** (3)
Solution:
Chilli has actinomorphic flowers whereas gulmohur, *Cassia* and bean have zygomorphic flowers.

(103) **Answer :** (2)
Hint:
A few millimetres above the root cap is the region of meristematic activity.
Solution:
Rapid elongation is evident in the cells of the elongation zone, which is present above the region of meristematic activity.

(104) **Answer :** (2)
Solution:
The arrangement of flowers on floral axis of plant is known as inflorescence.

(105) **Answer :** (4)
Hint:
When stamens are united into one bunch or one bundle, then it is monoadelphous condition.
Solution:
Monoadelphous stamens are found in China rose.
Diadelphous stamens are found in pea and polyadelphous stamens are found in *Citrus*.

(106) **Answer :** (1)
Hint:
Castor is a dicot plant.
Solution:
In castor seed, two cotyledons are present. Monocot seeds have one large and shield shaped cotyledon known as scutellum.

(107) **Answer :** (1)

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Solution: Tetradynamous condition (2+4) in stamens

(108) Answer : (2)
Hint:
A virus is larger than sub-viral agents/particles.
Solution:
Viroids are infectious RNA particles. They lack protein. They cause diseases in plants only.
Viruses have proteinaceous capsid.

(109) Answer : (3)
Hint:
In brinjal, hypogynous flowers are present.
Solution:
In brinjal, gynoecium occupies the highest position while other parts are situated below it. The ovary is said to be superior.

(110) Answer : (3)
Hint:
Adhesion is denoted by a line drawn above the symbols of the floral parts.
Solution:
Fusion is indicated by enclosing the figure within bracket. 'Br' stands for bracteate condition.

(111) Answer : (4)
Hint:
Lichens cannot tolerate air pollution.
Solution:
Lichens can grow in most inhospitable and uninhabited place but cannot tolerate air pollution especially due to SO_2, therefore these are considered as pollution indicators.

(112) Answer : (4)
Hint:
Root synthesises plant growth regulators.
Solution:
The food manufactured in leaves is transported to roots, fruits and organs of storage through stem.

(113) Answer : (3)
Hint:
Calyx is the outermost whorl of the flower and its members are called sepals.
Solution:
Corolla is the second whorl of the flower which is generally brightly coloured, have fragrance and makes the flower more attractive.

(114) Answer : (2)
Hint:
Fibrous root system is observed in wheat plant.
Solution:
In wheat plant, the primary root is short lived and replaced by a large number of roots and these roots originate from base of the stem.

(115) Answer : (1)
Solution:
Members of Fabaceae show papilionaceous corolla.
Gram (Fabaceae family) has papilionaceous corolla.

(116) Answer : (3)
Hint:
In lotus, carpels are not fused.
Solution:
Canna shows the presence of asymmetric flowers. *Alstonia* shows whorled type of phyllotaxy.

(117) Answer : (3)
Solution:
Most of the dinoflagellates are motile, biflagellated, photosynthetic marine protists.

(118) Answer : (1)
Solution:
Answer (1)
Perianth is a term for fused calyx and corolla.

(119) Answer : (2)
Hint:

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W.M. Stanley showed that viruses could be crystallised.
Solution:
T.O. Diener discovered a new infectious agent that was smaller than viruses.
M.W. Beijerinck called the infectious living fluid as – *Contagium vivum fluidum.*
Dmitri Ivanosky recognised certain microbes as causal organism of the mosaic disease of tobacco.

(120) Answer : (3)
Solution:
Double stranded DNA is usually present in bacteriophages.

(121) Answer : (4)
Solution:
Creutzfeldt-Jacob disease in humans is caused by prions.

(122) Answer : (4)
Hint:
Cuscuta is a parasitic plant.
Solution:
In imperfect fungi, reproduction is only by asexual spores known as conidia.

(123) Answer : (3)
Solution:
Slime moulds produce fruiting bodies during unfavourable conditions because the spores have extremely resistant true walls and survive for many year.

(124) Answer : (2)
Solution:
Puccinia is a pathogenic fungi. Pathogenic fungi are not edible.

(125) Answer : (2)
Solution:
Mycelium is aseptate and coenocytic in Phycomycetes. *Albugo* is a member of Phycomycetes.

(126) Answer : (4)
Solution:
Protozoans are unicellular heterotrophic organisms, and believed to be primitive relatives of animals.

(127) Answer : (3)
Solution:
Red dinoflagellates*, e.g., Gonyaulax,* undergo rapid multiplication that makes the sea appear red (red tides).

(128) Answer : (3)
Hint:
The organisms that are chief producer in the ocean are diatoms.
Solution:
Diatoms belongs to the kingdom Protista. They are unicellular eukaryotic organism. They are found in fresh water as well as in ocean.

(129) Answer : (1)
Solution:
Euglena has pellicle but not cell wall.

(130) Answer : (1)
Hint:
Amoeboid protozoans possess silica shells in some marine forms.
Solution:
(i) Ciliated protozoans have a definite region for ingestion (gullet) and egestion.
(ii) Mycoplasma infects both plants and animals.

(131) Answer : (2)
Hint:
Fungi which show only vegetative phase are known as imperfect fungi.
Solution:
Deuteromycetes is commonly known as imperfect fungi. *Albugo*, is a parasitic fungi on mustard. *Neurospora* is used extensively in biochemical and genetic work. *Ustilago* causes smut disease.

(132) Answer : (2)
Hint:
Viroids are the causative agents of potato spindle tuber disease.
Solution:

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RNA of the viroids has a low molecular weight.

(133) Answer : (3)
Solution:

Morels, *Aspergillus*] Ascomycetes

Bracket fungi] Basidiomycetes

Alternaria, *Colletotrichum*, *Trichoderma*] Deuteromycetes

(134) Answer : (4)
Hint:
Asexual reproduction takes place by spore formation in fungi.
Solution:
Fission, budding and fragmentation are methods of vegetative reproduction.

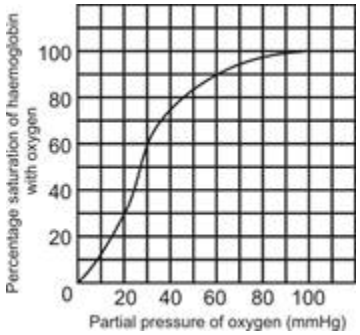
(135) Answer : (2)
Hint:
Members of ascomycetes are called sac fungi.
Solution:
Ascus is the site of karyogamy and meiosis in sac fungi.

ZOOLOGY

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(136) Answer : (1)
Solution:
Partial pressure of O_2 is most important factor in determining the saturation of haemoglobin.

(137) Answer : (1)
Solution:
A sigmoid curve is obtained when % saturation of Hb with O_2 is plotted against pO_2



(138) Answer : (1)
Solution:

Respiratory Gas	Atmospheric Air	Alveoli	Blood (Deoxygenated)	Blood (Oxygenated)	Tissues
pO_2	159	104	40	95	40
pCO_2	0. 3	40	45	40	45

(139) Answer : (1)
Solution:

Alveoli has high pO_2, low pCO_2, low H^+ and high pH.

(140) Answer : (2)
Solution:
About 7 per cent of CO_2 is carried in dissolved state through plasma.

(141) Answer : (3)
Hint:
Includes alveolar wall
Solution:
The diffusion membrane is made up of three major layers namely thin squamous epithelium of alveoli, the endothelium of alveolar blood capillaries and the basement substance in between them.

(142) Answer : (1)
Solution:
The pons region of the brain contains pneumotaxic centre that can moderate the functions of the respiratory rhythm centre. Neural signals from pneumotaxic centre can reduce the duration of inspiration and thereby alter the respiratory rate.

(143) Answer : (3)
Solution:
The membrane across which gas exchange takes place in the alveoli is extremely thin to aid in diffusion.

(144) Answer : (2)
Hint:
Fluid component of blood.
Solution:
Maximum amount of carbonic anhydrase is present in RBCs while minute quantities are present in fluid part of blood named plasma.

(145) Answer : (2)
Solution:
In occupational respiratory disorder, continuous exposure to harmful substances, gases, fumes and dust in the environment cause serious lung damage due to fibrosis of lungs.

(146) Answer : (1)
Solution:
About 4 mL of CO_2 is delivered by 100 mL of deoxygenated blood to the alveoli.

(147) Answer : (3)
Hint:
Medulla oblongata regulates respiration.
Solution:
Aortic arch and carotid artery are peripheral chemoreceptors for regulation of breathing. They can recognise change in CO_2 and H^+ concentration and send necessary signals to the respiratory rhythm centre (Medulla oblongata).

(148) Answer : (1)
Solution:
Increase in pCO_2 and hydrogen ion concentration activates chemosensitive area in medulla to increase breathing rate.

(149) Answer : (1)
Solution:
Exchange of gases between lungs and blood takes place by simple diffusion only. This process is driven by the difference in partial pressures of gases.

(150) Answer : (1)
Solution:
High pCO_2, low pH, high H^+ concentration and high temperature favour dissociation of O_2 from oxyhaemoglobin.

(151) Answer : (2)
Solution:
In emphysema, lungs remain inflated as exhalation becomes difficult.
In emphysema, alveolar sacs remain filled with air even after expiration.

(152) Answer : (3)
Solution:
The conducting part of the respiratory system is not involved in gaseous exchange.

(153) Answer : (1)

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Solution:
Chemosensitive area is highly sensitive to CO_2 and hydrogen ions.

(154) Answer : (1)
Solution:
Congestive heart failure is the state of heart when it is not pumping blood effectively enough to meet the needs of the body.

(155) Answer : (4)
Solution:
Fibroblasts form cells and matrix in all connective tissues except blood. Blood cells are formed within bone marrow.

(156) Answer : (2)
Solution:
Old, damaged RBCs are destroyed in the spleen and it is known as the RBC's graveyard. It is the secondary lymphoid organ also.

(157) Answer : (4)
Hint:
Basophils are least among leucocytes.
Solution:
Neutrophils - 60-65% of total WBCs
Lymphocytes - 20-25% of total WBCs
Monocytes - 6-8% of total WBCs
Eosinophils - 2-3% of total WBCs
Basophils - 0.5-1% of total WBCs

(158) Answer : (3)
Solution:
People with 'AB' blood group are called universal recipients. Individuals with 'O' blood group are called universal donor.

(159) Answer : (3)
Hint:
Lymphatic fluid mixes with deoxygenated blood.
Solution:
An elaborate network of vessels called the lymphatic system collects this fluid and drains it back to the major veins.

(160) Answer : (4)
Hint:
Thrombokinase catalyses conversion of prothrombin to thrombin.
Solution:

$$\text{Fibrinogen} \xrightarrow{\text{Thrombin}} \text{Fibrin}$$

(161) Answer : (4)
Hint:
Identify a body fluid that cannot clot.
Solution:
Plasma without clotting factor is called serum.

(162) Answer : (3)
Hint:
Mother is Rh negative in this condition.
Solution:
Injecting Rh antigen to Rh -ve mother's blood leads to agglutination reactions. In this condition, anti-Rh antibodies are administered to the mother immediately after the delivery of the first Rh +ve child.

(163) Answer : (1)
Hint:
Atrioventricular valves.
Solution:
First heart sound occurs due to closure of AV valves *i.e.*, bicuspid and tricuspid valves. Second heart sound occurs due to closure of semilunar valves.

(164) Answer : (2)
Hint:
Erythrocytes, leucocytes and platelets are collectively called formed elements.
Solution:
Platelets are also called thrombocytes. They are cell fragments produced from megakaryocytes. They release a variety of substances most of which are involved in the coagulation or clotting of blood.

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(165) Answer : (2)
Hint:
Pulmonary artery and aorta possess same type of valves.
Solution:
The openings of right and left ventricles into the pulmonary artery and aorta respectively are provided with the semilunar valve. Bicuspid or mitral valve guards the opening between the left atrium and the left ventricle.

(166) Answer : (4)
Solution:
Fibrinogens are required for clotting or coagulation of blood. Globulins are primarily involved in defense mechanisms of the body and albumins help in maintaining osmotic balance.

(167) Answer : (2)
Solution:
Parasympathetic nervous system decreases cardiac output as it releases Ach which slows down impulse conduction in cardiac muscle.

(168) Answer : (3)
Solution:
Crocodiles are reptiles with completely divided four chambered heart. Amphibians have three chambered heart.

(169) Answer : (4)
Solution:
Left atrium and right ventricle are included in pulmonary circulation.

(170) Answer : (1)
Hint:
Depolarisation of ventricles leads to contraction of ventricles.
Solution:
QRS wave represents the ventricular depolarisation in ECG. After the initiation of ventricular depolarisation, ventricular contraction starts and is completed with the end of T-wave.

(171) Answer : (3)
Hint:
Pressure within ventricles decreases.
Solution:
During ventricular diastole, the ventricular pressure falls causing the closure of semilunar valves, which prevent the backflow of blood into ventricles from aorta and pulmonary artery.

(172) Answer : (4)
Solution:
All the four chambers of heart are in a relaxed state*, i.e.,* they are in joint diastole. As the tricuspid and bicuspid valves are open, blood from the pulmonary veins and vena cava flows into the left and the right ventricles, respectively, through the respective left and right atria. The semilunar valves are closed at this stage. The SAN now generates an action potential which stimulates both the atria to undergo a simultaneous contraction, known as the atrial systole. This increases the flow of blood into the ventricles by about 30 per cent.

(173) Answer : (3)
Solution:
Gaseous exchange takes place in the alveoli of lungs. There are 2 pairs of pulmonary veins connecting to the left atrium of heart through which oxygenated blood from lungs enter.

(174) Answer : (2)
Solution:
Answer (2)

RBC count is 5.5 million/mm^3, WBC count 6000-8000/mm^3 and platelets are 1.5 – 3.5 lakh/mm^3.

(175) Answer : (1)
Solution:
The Bundle of His is the conducting part of cardiac musculature that passes through the interventricular septum. , Bundle of His helps to conduct impulses from the atrioventricular node to the ventricles.

(176) Answer : (2)
Hint:
Cardiac output = Stroke volume × heart rate
Solution:
Cardiac output = 4900 mL
Blood volume in ventricles at the end of diastole = 140 mL
Blood volume in ventricles at the end of systole = 70 mL

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Stroke volume = 140 – 70 = 70 mL
So, 4900 mL = 70 mL × heart rate
Heart rate = 70 beats per minute.

(177) Answer : (2)
Solution:
Blood moves out of the ventricles into the respective vessels during ventricular systole and atrio-ventricular valves are closed during ventricular systole.

(178) Answer : (3)
Solution:
Tunica media is the middle layer of blood vessels made up of smooth muscle and elastic fibres. Smooth muscle fibres are uninucleated, spindle shaped(fusiform) and involuntary in nature.

(179) Answer : (2)
Solution:
A thin, muscular wall called the inter-atrial septum separates the right and the left atria, whereas a thick-walled, the inter-ventricular septum, separates the left and the right ventricles. The atrium and the ventricle of the same side are also separated by a thick fibrous tissue called the atrio-ventricular septum.

(180) Answer : (2)
Hint:
They are the least abundant WBCs present in human blood.
Solution:
Basophils are the least abundant granulocytes. They can secrete heparin, histamine and serotonin and are involved in inflammatory reactions.
• Eosinophils (2-3 per-cent of total WBCs) resist infections and are associated with allergic reactions.
• Neutrophils (60-65 per cent of total WBCs) and monocytes (6-8 per cent of total WBCs) are phagocytic in nature.

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